

Microbes & cells — overview

Body cells and pathogens at a glance

Variant: basics

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Cell categories follow the 'Types of cells' chart. All illustrations are taken from the public-domain NIH BioArt collection (bioart.niaid.nih.gov).

Stem cells are unspecialised cells that self-renew and can develop into different mature cell types. Embryonic and induced pluripotent stem cells (iPS) can become almost anything. Tissue-specific stem cells supply fresh cells to blood, skin, gut, nerves and bone.

Embryonic stem cell (ESC)

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FUNCTION: Pluripotent cell from the early embryo (blastocyst). Can develop into any of the body's 200+ cell types.

DEPENDS ON / TARGETS: Differentiated in vitro into a target lineage with growth factors. Starting point for many regenerative therapies.

Induced pluripotent stem cell (iPS)

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FUNCTION: A body cell (e.g. skin) reset by a handful of transcription factors into an embryonic-like state — can develop into almost any cell type.

DEPENDS ON / TARGETS: Differentiated in vitro into a target lineage with specific growth factors. Source for many therapies.

Hematopoietic stem cell (HSC)

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FUNCTION: Mother of all blood cells. Lives in bone marrow, divides rarely and gives rise to every blood lineage — from red blood cells to lymphocytes.

DEPENDS ON / TARGETS: Needs niche cells in bone marrow (osteoblasts, stromal cells) and growth factors.

Mesenchymal stem cell (MSC)

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FUNCTION: Reserve cell for supporting tissues. Can turn into bone, cartilage, muscle, fat or connective-tissue cells.

DEPENDS ON / TARGETS: Lives in bone marrow, fat tissue and umbilical cord. Supplies precursors on demand.

Neural stem cell (NSC)

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FUNCTION: Precursor of all neural cell types. In the adult brain only active in a few niches (e.g. hippocampus).

DEPENDS ON / TARGETS: Responds to growth factors (EGF, FGF) and tissue cues. Supplies neurons and glial cells.

Endothelial progenitor cell (EPC)

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FUNCTION: Young endothelial cell. Mobilised during wound healing and angiogenesis to replace damaged vessel wall.

DEPENDS ON / TARGETS: Recruited from bone marrow; responds to VEGF and other vascular growth factors.

Epithelial cells line every inner and outer surface of the body: skin, the mucosa of mouth, GI tract and lung, plus glands. They form the body's first mechanical and biochemical barrier against the environment.

Keratinocyte (skin cell)

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FUNCTION: Building block of the outer skin. Lays down keratin, cornifies upward and forms a waterproof, robust barrier.

DEPENDS ON / TARGETS: Renewed from stem cells in the basal layer. Cooperates with melanocytes and Langerhans cells.

Enterocyte (gut cell)

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FUNCTION: Tall column-shaped cell with a dense brush border (microvilli). Absorbs nutrients from the gut and passes them on.

DEPENDS ON / TARGETS: Replaced every few days by intestinal stem cells. Forms a barrier against gut microbes.

Goblet cell

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FUNCTION: Goblet-shaped mucus producer in gut and airways. Releases mucin packets that form a protective gel over the epithelium.

DEPENDS ON / TARGETS: Cooperates with enterocytes. Protects against dryness and bacterial contact.

Paneth cell

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FUNCTION: Sits at the base of intestinal crypts and fires antimicrobial proteins (defensins, lysozyme). Protects the intestinal stem cells.

DEPENDS ON / TARGETS: Together with stem cells forms the crypt-base niche.

Alveolar cell type II

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FUNCTION: Small cuboidal lung cell. Makes surfactant — the film that keeps alveoli open. Also acts as stem cell for type I cells.

DEPENDS ON / TARGETS: Lives among the thin type-I cells that handle gas exchange.

Urothelial cell

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FUNCTION: Stretchable cell of the bladder wall. Reshapes its surface as the bladder fills and seals against acidic urine.

DEPENDS ON / TARGETS: Forms a barrier between urine and body. Renewed from basal cells.

Nerve cells and their support team (glia) build the entire nervous system — brain, spinal cord and the peripheral nerves that supply muscles, skin and organs. They communicate through electrical impulses and chemical messengers (neurotransmitters).

Neuron

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FUNCTION: Compute-and-signal cell: collects inputs through dendrites, sums them up and fires electrical impulses down its axon.

DEPENDS ON / TARGETS: Depends on astrocytes (support) and oligodendrocytes (myelin). Communicates through synapses with other neurons.

Motor neuron

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FUNCTION: Nerve cell that carries commands from the spinal cord to the muscles. Its axon can be a meter long and ends at the motor end plate.

DEPENDS ON / TARGETS: Drives skeletal muscle via acetylcholine. In ALS these are the cells that die off.

Astrocyte

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FUNCTION: Star-shaped support cell. Feeds neurons, balances ions and neurotransmitters, helps form parts of the blood-brain barrier.

DEPENDS ON / TARGETS: Supports neurons and regulates local blood flow.

Oligodendrocyte

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FUNCTION: Wraps several axons at once with myelin in brain and spinal cord — the electrical insulation of nerves.

DEPENDS ON / TARGETS: Partly fuels axons. In multiple sclerosis this myelin layer is the target.

Microglia

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FUNCTION: Brain-resident macrophages: constantly probe the tissue with thin processes, clear debris, prune synapses.

DEPENDS ON / TARGETS: Yolk-sac derived in the embryo. First line of defence against brain infections.

Schwann cell

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FUNCTION: Myelin maker in the peripheral nervous system: each Schwann cell wraps one segment of a single axon.

DEPENDS ON / TARGETS: Also helps peripheral nerves regenerate after injury.

Reproductive (germ) cells pass DNA to the next generation. Egg and sperm fuse at fertilisation. Sertoli, Leydig, granulosa and theca cells nourish, protect and hormonally steer the maturing germ cells.

Spermatozoon (sperm)

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FUNCTION: Male germ cell with a half set of chromosomes. Swims actively to the egg and delivers the father's DNA at fertilisation.

DEPENDS ON / TARGETS: Matures in the testis under the guidance of Sertoli cells. Needs testosterone from Leydig cells.

Oocyte (egg cell)

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FUNCTION: Largest cell in the body, loaded with nutrients and maternal factors. Matures once per cycle and waits for the sperm.

DEPENDS ON / TARGETS: Nourished by granulosa and theca cells inside the follicle. After fertilisation, the embryo develops.

Sertoli cell

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FUNCTION: Nurse cell for sperm: sits in the testicular tubules, feeds maturing sperm and shields them from the immune system (blood-testis barrier).

DEPENDS ON / TARGETS: Responds to FSH from the pituitary and steers sperm maturation via inhibin and androgen-binding protein.

Leydig cell

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FUNCTION: Hormone factory of the testis: makes testosterone on orders from the pituitary. Drives sperm production and male traits.

DEPENDS ON / TARGETS: Responds to LH from the pituitary. Supplies Sertoli cells and the whole body with testosterone.

Granulosa cell

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FUNCTION: Wrapper cell in the ovarian follicle: feeds the maturing egg and converts androgens from theca cells into estrogen.

DEPENDS ON / TARGETS: Responds to FSH. After ovulation, with theca cells, it forms the corpus luteum.

Theca cell

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FUNCTION: Outer layer of the ovarian follicle. Produces androgens that granulosa cells convert into estrogen.

DEPENDS ON / TARGETS: Responds to LH from the pituitary. Works closely with granulosa cells to balance hormones across the cycle.

Bone is a living tissue that is constantly built and broken down. Osteoblasts lay down new matrix, osteoclasts dissolve old bone, osteocytes monitor the balance. Cartilage, tendons and connective tissue complete the musculoskeletal system.

Osteoblast (bone builder)

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FUNCTION: Builds new bone: secretes collagen and mineralises it with calcium. Walls itself in and becomes an osteocyte.

DEPENDS ON / TARGETS: Balances with osteoclasts that resorb bone. Hormonally controlled by PTH and vitamin D.

Osteoclast

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FUNCTION: Multi-nucleated giant cell that resorbs bone. Secretes acid and enzymes that dissolve calcium and matrix.

DEPENDS ON / TARGETS: Comes from the monocyte lineage. Balances osteoblasts — in osteoporosis the osteoclasts win.

Osteocyte

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FUNCTION: Mature bone cell, walled inside its own matrix. Senses mechanical load and signals osteoblasts and osteoclasts when to remodel.

DEPENDS ON / TARGETS: Connects to neighbours through long cytoplasmic processes in tiny canals. Lives for decades.

Chondrocyte (cartilage cell)

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FUNCTION: Sits alone or in small groups inside a 'lacuna' of cartilage matrix. Produces and maintains the rubbery cartilage tissue.

DEPENDS ON / TARGETS: Cartilage has almost no blood supply — cells rely on diffusion. Crucial for joints, intervertebral discs, growth plates.

Tenocyte (tendon cell)

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FUNCTION: Slim, in-line cell of tendons. Produces strong collagen fibres that anchor muscle to bone.

DEPENDS ON / TARGETS: Responds to mechanical tension. Heals slowly because tendons have poor blood supply.

Fibroblast

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FUNCTION: Connective-tissue all-rounder: makes collagen and other fibres, keeps skin and organs in shape and closes wounds.

DEPENDS ON / TARGETS: Activated at wounds by growth factors (TGF- β , PDGF). Works closely with macrophages.

Fat cells (adipocytes) store energy as lipid and also act as an endocrine gland (leptin, adiponectin). White fat is the classic energy store, brown fat burns it to make heat, beige fat is in between. Precursors and stem cells keep the supply going.

White adipocyte

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FUNCTION: Classic fat cell: a single huge lipid droplet pushes the nucleus and cytoplasm to the rim. Stores energy for lean times.

DEPENDS ON / TARGETS: Stimulated by insulin to take up fat. Releases leptin, which tells the brain about energy stores.

Brown adipocyte

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FUNCTION: Body heater: many small lipid droplets and unusually many mitochondria. Burns fat directly into heat (via UCP1) rather than ATP.

DEPENDS ON / TARGETS: Abundant in babies, in adults mainly around neck and collarbone. Activated by cold.

Beige adipocyte

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FUNCTION: In-between form: lives in white fat depots but can be switched on by cold or exercise to a heat-producing state ('browning').

DEPENDS ON / TARGETS: Responds to irisin (from muscle) and noradrenaline. A therapeutic target against obesity.

Preadipocyte

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FUNCTION: Direct precursor of the fat cell. Waits as a spindle-shaped cell in adipose tissue and matures into an adipocyte on demand.

DEPENDS ON / TARGETS: Responds to hormonal cues (insulin, glucocorticoids, PPAR γ). Provides new fat cells during weight gain.

Lipoblast

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FUNCTION: Early precursor from a mesenchymal stem cell. Picks up the first small lipid droplets before maturing into a preadipocyte.

DEPENDS ON / TARGETS: Key in embryonic fat development; reappear in liposarcomas.

Adipogenic progenitor

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FUNCTION: A mesenchymal-stem-cell-derived precursor that can still take several routes — most often becomes a fat cell.

DEPENDS ON / TARGETS: Lives in the vascular niche of adipose tissue. Responds to nutrition, hormones and inflammatory signals.

Red blood cells (erythrocytes) carry oxygen from the lungs into tissue and CO_2 back. They mature in bone marrow in stages — from erythroblast through reticulocyte to the anucleate erythrocyte. Megakaryocytes and platelets are close cousins from the same precursor.

Erythrocyte (red blood cell)

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FUNCTION: Oxygen carrier: no nucleus, packed with haemoglobin. Lives about 120 days and circulates millions of times.

DEPENDS ON / TARGETS: Production driven by erythropoietin from the kidney. Delivers oxygen to every body cell.

Reticulocyte

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FUNCTION: Young red blood cell: just released from bone marrow, still carrying RNA remnants. Matures into a full erythrocyte in one to two days.

DEPENDS ON / TARGETS: Blood share (~1 %) shows how fast bone marrow is making new cells — an important clinical marker.

Erythroblast

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FUNCTION: Nucleated precursor in bone marrow. Builds up haemoglobin, finally ejects its nucleus and becomes a reticulocyte.

DEPENDS ON / TARGETS: Matures through several stages (proerythroblast → basophilic → polychromatic → orthochromatic).

Megakaryocyte

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FUNCTION: Giant bone-marrow cell with a multi-lobed nucleus. Pinches off thousands of platelets — a cousin of the erythrocyte lineage.

DEPENDS ON / TARGETS: Driven by thrombopoietin (TPO) from the liver. Shares the MEP precursor with the erythrocyte.

Thrombocyte (platelet)

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FUNCTION: Tiny anucleate fragments. Plug wounds, form the first clot and trigger the coagulation cascade.

DEPENDS ON / TARGETS: Originate from megakaryocytes. Cooperate closely with coagulation factors and endothelial cells.

Sickle cell (deformed erythrocyte)

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FUNCTION: Pathological erythrocyte form: mutated haemoglobin (HbS) deforms the cell into a sickle when oxygen drops. Clogs capillaries and causes sickle-cell anaemia.

DEPENDS ON / TARGETS: Autosomal recessive. Carriers of one copy are partly protected from malaria — hence common in endemic regions.

Immune cells recognise, tag and dispose of pathogens and rogue cells. T and B cells are the specialised troops of adaptive immunity, macrophages and neutrophils the rapid-response squad of innate immunity, NK cells the bouncers against virus-infected and tumour cells.

Helper T cell (CD4)

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FUNCTION: Conductor of the immune response: recognises antigen fragments shown by other cells and releases messengers that switch other cells on.

DEPENDS ON / TARGETS: Needs antigen-presenting cells (dendritic cells, macrophages). Activates B cells and CD8 killer cells.

Cytotoxic T cell (CD8)

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FUNCTION: Killer cell: spots infected or tumour cells by their MHC-I signal and kills them precisely with perforin and granzymes.

DEPENDS ON / TARGETS: Primed by dendritic cells, often with help from CD4 T cells. Targets virus-infected body cells and tumour cells.

B cell

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FUNCTION: Antibody factory: binds antigen with its B-cell receptor and matures into plasma cells that pump out tailored antibodies.

DEPENDS ON / TARGETS: Usually needs help from CD4 T cells to switch into high gear. Targets free bacteria, virus particles and toxins in blood and tissue.

Natural killer cell (NK)

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FUNCTION: Innate-immune bouncer: kills cells that have lost their MHC-I 'ID badge' — typical for virus infection or cancer.

DEPENDS ON / TARGETS: Revved up by interferons and IL-12. Kills virus-infected and tumour cells without prior priming.

Macrophage

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FUNCTION: 'Big eater': engulfs bacteria, dead cells and debris, presents fragments to T cells and calls reinforcements with cytokines.

DEPENDS ON / TARGETS: Differentiates from blood monocytes. Phagocytoses bacteria, fungi, cancer cells and cleans up wounds.

Neutrophil

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FUNCTION: First responder: arrives first at sites of inflammation, eats bacteria, dumps antimicrobials and even casts sticky DNA nets (NETs).

DEPENDS ON / TARGETS: Chemokine-guided. Targets bacteria and fungi; lives only hours to a few days.

Pathogens are anything that can cause disease: bacteria, viruses, fungi, parasites and prions. They live in air, water, soil, animals and on our skin — most microbes are harmless or beneficial, a few are highly dangerous.

Cocci (round bacteria)

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FUNCTION: Round bacteria, often in clusters or chains (Staphylococci, Streptococci). Some are harmless guests, others cause pneumonia, skin or throat infections.

DEPENDS ON / TARGETS: Phagocytosed by neutrophils and macrophages; antibodies help tag them.

Rod bacteria

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FUNCTION: Elongated bacteria like E. coli, Salmonella, tuberculosis bacilli. Multiply fast and can release toxins.

DEPENDS ON / TARGETS: Phagocytosed; intracellular species (TB) need T-cell help to be cleared.

Virus

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FUNCTION: Not really alive: just genetic material in a shell. Sneaks into body cells and forces them to build new virus particles.

DEPENDS ON / TARGETS: Hunted by CD8 T cells and NK cells; antibodies bind free virus particles.

Fungus

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FUNCTION: Eukaryotic microbes — yeasts or filaments (hyphae). Examples: Candida in mouth/gut, Aspergillus in lungs.

DEPENDS ON / TARGETS: Controlled by neutrophils, macrophages and Th17 cells.

Parasite

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FUNCTION: From single-celled (Plasmodium → malaria) to worms. Lives in or on its host and damages it in the process.

DEPENDS ON / TARGETS: Often fought with IgE, eosinophils and mast cells; antibodies and Th2 responses are central.

Prion

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FUNCTION: Not alive, not a virus — a misfolded protein that forces other proteins to misfold the same way. Causes diseases like Creutzfeldt-Jakob, BSE.

DEPENDS ON / TARGETS: Largely invisible to the immune system; no antibody defence.

Bacteria are single-celled organisms without a true nucleus (prokaryotes). They live everywhere — in soil and water, on skin and in the gut. Most are harmless or beneficial; a few cause millions of cases of disease worldwide every year.

Mycobacterium tuberculosis (TB)

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FUNCTION: Waxy-coated rod that hides inside macrophages and can lurk in the lungs for years. Causes tuberculosis — still one of the deadliest infections worldwide.

DEPENDS ON / TARGETS: Airborne transmission. Contained by macrophages and CD4 T cells; without them the bacterium persists.

Staphylococcus aureus (MRSA)

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FUNCTION: Grape-cluster cocci. Often harmless on skin and in the nose, but can cause wound infections, abscesses and sepsis. MRSA strains resist many antibiotics.

DEPENDS ON / TARGETS: Phagocytosed by neutrophils and tagged by antibodies. Hospital hygiene is key against resistance spread.

Streptococcus pneumoniae

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FUNCTION: Cocci in chains, often in the nasopharynx. Leading cause of pneumonia, otitis media and meningitis, especially in children and the elderly.

DEPENDS ON / TARGETS: Polysaccharide capsule blocks phagocytosis; antibodies and complement break through. Vaccines available.

Escherichia coli

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FUNCTION: Gut rod bacterium. Most strains are harmless or useful; some (e.g. EHEC) produce toxins and cause severe diarrhoea and kidney failure.

DEPENDS ON / TARGETS: Spread by faecal-oral route and contaminated food. Controlled in the gut by IgA antibodies and the mucus barrier.

Salmonella enterica

Image missing

FUNCTION: Rod bacterium from many animals' guts. Picked up via contaminated eggs, poultry or water, it causes diarrhoea (salmonellosis) or typhoid fever.

DEPENDS ON / TARGETS: Cleared by neutrophils, macrophages and the gut epithelium; Th1 responses are central to elimination.

Helicobacter pylori

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FUNCTION: Spiral bacterium that survives in stomach acid. Causes gastritis, peptic ulcers and raises the risk of stomach cancer.

DEPENDS ON / TARGETS: Usually transmitted in childhood. Treated with antibiotics plus acid blockers; vaccine still in development.

Viruses are genetic material in a coat and need a host cell to replicate. Fungi and parasites, by contrast, are organisms in their own right. The pathogens shown here infect humans worldwide, ranging from harmless colds to severe pandemics.

Influenza virus (flu)

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FUNCTION: Respiratory RNA virus with a segmented genome. Mutates constantly, so vaccines are reformulated each year. Causes seasonal waves and occasional pandemics.

DEPENDS ON / TARGETS: Spread by droplets. CD8 T cells and antibodies clear infected cells and free virus particles.

SARS-CoV-2 (COVID-19)

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FUNCTION: Coronavirus with a 'crown' of spike proteins. Binds ACE2 receptors in the airways and triggered the global COVID-19 pandemic in 2020.

DEPENDS ON / TARGETS: Spreads via aerosols. mRNA and vector vaccines train antibodies and T cells against the spike protein.

HIV

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FUNCTION: Retrovirus that targets CD4 helper T cells and integrates its genome into human DNA. Untreated, it destroys the immune system (AIDS).

DEPENDS ON / TARGETS: Spread via blood, sexual contact and mother-to-child. Antiretroviral therapy (ART) keeps the virus suppressed long-term.

Hepatitis B virus (HBV)

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FUNCTION: DNA virus that infects the liver and persists inside liver cells. Chronic infection can lead to liver cirrhosis and liver cancer.

DEPENDS ON / TARGETS: Spread via blood, sexual contact and perinatally. Vaccine available and included in many national immunisation programmes.

Plasmodium (malaria)

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FUNCTION: Single-celled parasite spread by Anopheles mosquitoes. First infects the liver, then red blood cells, triggering cycles of fever. Kills over 500,000 people per year worldwide.

DEPENDS ON / TARGETS: Treated with antimalarial drugs; new vaccines (RTS,S, R21) now being rolled out in endemic regions.

Candida albicans

Image missing

FUNCTION: Yeast that lives harmlessly in many people's mouths, gut and genital tract. With a weakened immune system or after antibiotics it can spread and cause serious infections.

DEPENDS ON / TARGETS: Kept in check by Th17 cells, neutrophils and the mucosal microbiome.

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Microglia — NIH BioArt *Public domain (NIH BioArt)*

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